Vineeth Raghav Gururajan

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PROFILE

M.Sc. Electrical Engineering student · Python Software Development · CAD (AutoCAD, CATIA, NX, Altium) · Simulation & Automation

- 1 year of industry experience in electronics design, with Python and embedded C automation of testing and performance analysis
- CAD experience with AutoCAD, CATIA, NX and Altium Designer; built a battery simulation model with statistical parameter sampling
- Enrolled at the University of Stuttgart; available from 15 October 2026, up to 20 h/week, fully remote

EXPERIENCE

Electronic Design Intern – FocalSystech Pvt. Ltd., Chennai, India 07/2024 – 07/2025

- **Python automation:** Automated testing and performance analysis in embedded C and Python, reducing manual test effort
- **Circuit & PCB design:** Designed and validated electronic circuits; created PCB layouts in Altium and supported prototype fabrication, bring-up and validation of the assembled boards
- **System integration:** Defined electrical and hardware–software interfaces between sensors, actuators and microcontrollers (CAN, I2C, SPI, UART) for reliable operation in IoT devices
- **Component evaluation & selection:** Evaluated and selected electronic components against technical requirements for embedded systems, working from schematics and datasheets

EDUCATION

M.Sc. Electrical Engineering – University of Stuttgart 10/2025 – expected 2027

- Intelligent Power Grids, Power Systems Engineering, Battery Modeling, Power Electronics, Control Systems

B.Tech. Electronics and Computer Engineering – SRM Institute of Science and Technology, Chennai · 2021 – 2025
GPA 8.7/10

PROJECTS

Battery Cell Modeling – Team project, University of Stuttgart 2026

- Analysed cell-to-cell variance caused by electrical, thermal, aging and manufacturing tolerances by simulation; implemented and debugged statistical parameter sampling in the model. Directly relevant to battery pack and battery management design

64-bit RoBA Multiplier with MAC Unit – B.Tech. final-year project, Verilog 2025

- Replaced the Kogge–Stone adder with an area- and power-efficient Han–Carlson adder; synthesised and validated the design for improved speed, power and area. An optimisation exercise: same function, lower cost and complexity

Duplicate Question Detection – Python, NLP

- Built a Siamese neural network for semantic similarity detection using Word2Vec embeddings

SKILLS

CAD	AutoCAD, CATIA, NX; Altium Designer (PCB layout)
Modeling & Simulation	MATLAB, circuit simulation (PSpice), battery cell modeling, statistical parameter sampling and tolerance analysis in simulation
Systems Engineering	Requirements-based design and validation, interface definition between sensors, actuators and controllers, system integration, debugging
Power & E-Mobility	Power electronics, battery modeling, control systems, sensors and actuators
Programming	Python, C/C++, MATLAB, Verilog; microcontrollers (ARM, ESP32, Arduino)
Integration & Test	CAN, I2C, SPI, UART, system integration, test automation, debugging

LANGUAGES & OTHER

English: fluent · **German:** A2, preparing for the telc B1 exam
Certificates: PSpice Fundamentals (TVS India), Certificate of Training (Livewire Ltd.) · **Volunteering:** STEM workshops for underprivileged students; post-cyclone relief in Chennai